01 CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

PAGE / 08

Real 12:1 Planetary Gears + Pie-Slice Time Wedges

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • COVER (TOP-LEFT)

WHAT IS THIS BOX FOR? A CLOCK THAT ACTUALLY TELLS THE TRUTH.

Most toy clocks have loose hands that lie to you. Ours has real planetary gears inside: spin the minute ring 360° and the hour hand moves 1 hour!

REAL 12:1 PLANETARY GEARS INSIDE

Why is reading an analog clock tricky at 2:45? Because the hour hand is almost on the 3! Our geared clock links both hands mechanically:

- **1 Full Turn of Minute Ring (360°) = 30° Hour Hand Turn (1 Hour)**
- **Always In Sync: You can never put the hour hand in the wrong place!**
- Drop-In Pie Wedges: 15-minute sectors show quarter-hours as fractions.

See continuous gear motion and fraction slices at the exact same time!

ANALOG-TO-DIGITAL PRACTICE LID

The box lid has digital clock slots on top! Snap in the minute blocks (:00, :15, :30, :45) and AM/PM tiles to match your analog dial.

WHAT'S INSIDE YOUR BOX (19 PIECES)

- 1× Geared Planetary Clock Core (12:1 Ratio) + Hour & Minute Hands
- 4× Quarter-Hour Pie Wedges (:15, :30, :45, :60) + 12× Digital & AM/PM Tiles
- Keep this folded guide inside your box lid for examples & part IDs!

★ YOUR FIRST 60 SECONDS WITH THIS BOX — START HERE!

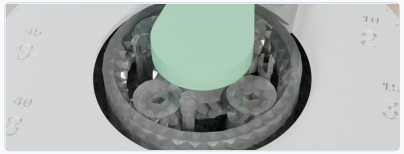
1. **SPIN THE MINUTE RING ONE FULL TURN:** Turn the Green minute hand 360° from 12 back to 12–watch the Red hour hand advance 1 hour (30°)!
2. **DROP PIE WEDGES ONTO THE FACE:** Lay two 15-minute wedges from 12 to 6 to cover half the circle (30 minutes = 1/2 hour).
3. **PRESS THE BOX ICON KEY [🕒] / [g][▸HMS]:** Type 3.30 [f][▸HR] 0.45 [f][▸HR] [+] [g][▸HMS] on your calculator to bridge across 4:00 to 4:15!

BOX ICON = CALCULATOR [🕒] CLOCK MODE & [G][▸HMS] TIME KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([🕒] / [g][▸HMS] / [f][▸HR])! Build every lesson with your blocks, then press the key.

12:1 PLANETARY GEARS ↔ BOX [🕒] & [g][▸HMS] ICON

STEP A: HANDS-ON WITH EPICYCLIC GEARS — 1:1 CAD SCENE



Minute Turn	Hour Advance	Box Icon Key
360° = 60 Minutes	30° = 1 Hour	[🕒] & [g][▸HMS]

▲ **Spin the Green minute hand and watch the internal planet gears** drive the Red hour hand at a true 12:1 ratio!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

FACTOR TREE: 60 MINUTES = 2 × 2 × 3 × 5

60 MIN (360° TURN)

60 [g][FACT] | ÷ 4 = 15m Wedge

15m (1/4) | 15m (1/4) | 15m (1/4) | 15m (1/4)

60 = 2²×3×5 • 60 ÷ 4 = 15 min

PRESS BOX ICON & RPN KEYS (SCG):

1. FACTOR 60: 6 0 g FACT 2²×3×5
2. 1/4 WEDGE: 4 ÷ = 15 min

✓ DISPLAY (X) = 15

KEYSTROKES: 60 [g] [FACT] 4 [÷]

Factor 60 minutes (2²×3×5) and divide by 4 quadrants to get each 15-minute wedge!

WELCOME TO HANDS-ON MATH • KEEP THIS GUIDE IN YOUR BOX LID

STEP 1: UNFOLD RIGHT → PAGE 02

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

02 HOW IT WORKS: THE CORE IDEA

PAGE / 08

12:1 Geared Hand Link & Modular-12 Clock Arithmetic

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 1ST UNFOLD (TOP-RIGHT)

HOW PLANETARY GEARS LINK ANGLES TO TIME

Inside your clock is an epicyclic ("planetary") gear train that divides every rotation of the outer minute ring by 12:

Gear Ratio = ω_minute / ω_hour = 12 : 1

The angle of the hour hand θ_hour is always locked to 1/12th of the angle swept by the minute hand θ_minute:

θ_hour = (1 / 12) × θ_minute

INSIDE THE PLANETARY GEARBOX

- Sun Gear: 12-tooth center driver | Planet Gears: 18-tooth orbiting idlers
- Ring Gear: 48-tooth outer ring with smooth finger-turn grip
- Continuous Mesh: 1 full turn (360°) advances the hour hand by 30°
- Built to Last: Gears are permanently meshed so they never slip out of time

TIME IS JUST FRACTIONS OF A CIRCLE

Why do we call 15 minutes "a quarter past" and 30 minutes "half past"? Drop the colored quadrant wedges onto the clock face and you'll see: 15 minutes is a 90° quarter-circle (1/4), and 30 minutes is a 180° half-circle (1/2)!

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE — VISUAL PROOF

✗ **TRAP 1 – Misreading 2:45 as '3:45':**
✓ **FIX:** Because the Red hour hand is geared to move gradually toward 3, learners see WHY it's almost 3:00.

✗ **TRAP 2 – Adding time like base-10 (1:45 + :30 ≠ 1:75):**
✓ **FIX:** Putting two :15 wedges past 1:45 (or pressing [g][▸HMS]) visibly crosses 12:00 into 2:15!

✗ **TRAP 3 – Not getting 'quarter past' or 'quarter to':**
✓ **FIX:** Each 15-minute wedge is literally 1/4 (one quarter) of the circular clock face.

CALCULATOR CHECK: 60 [g] [FACT] 4 [÷] = X = 15

TIME TO ANGLE REFERENCE TABLE

INTERVAL	MINUTE ARC	HOUR ARC	FRACTION
60 Min (1 Hr)	360° (Full)	30.0°	1 / 1 (Whole)
45 Min (3/4)	270°	22.5°	3 / 4 Circle
30 Min (1/2)	180°	15.0°	1 / 2 Circle
15 Min (1/4)	90°	7.5°	1 / 4 Circle
5 Min (1/12)	30°	2.5°	1 / 12 Circle
1 Minute	6°	0.5°	1 / 60 Circle

WHY YOUR BRAIN LOVES THIS

- No Batteries Needed: Powered 100% by your fingers and real gears.
- Zero Loose Screws: Monolithic gear assembly can't come apart in a desk.
- No Base-60 Headaches: Count elapsed time by dropping in 15-min wedges.
- Glow-in-the-Dark Hands: High-contrast hands are easy to read anywhere.
- Easy Care: Wipe clean or rinse in cool soapy water (never oil the gears).

1:1 CAD REFERENCE VIEW — BOX ICON & FORMULA QUICK-REFERENCE

1:1 BLOCK CAD ANATOMY

SOCKET (1.85mm) ▲ 1.85m

1U = 7.00 mm VISIBLE DATUM

7mm

BASE PLUG (1.85mm) ▼ 1.85m

✓ FLUSH PLUG-AND-SOCKET JOINT

Socket swallows plug completely: **Only 7.00mm true heights add!**

Box Button Icon Key: [🕒] Clock Mode & [g][▸HMS] / [f][▸HR]

Planetary Gear Law: θ_hour = θ_minute ÷ 12

1 Hour Advance: 360° minute turn = 30° hour turn

15-Min Quadrant Wedge: 1/4 Hour = 90° = π/2 radians

Add HH.MM Times: T₃ [f][▸HR] T₂ [f][▸HR] [+] [g][▸HMS]

REAL PLANETARY GEARS • THE HANDS CAN NEVER FALL OUT OF SYNC

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

03 EXAMPLE 1: HANDS-ON WALKTHROUGH

PAGE / 08

Spin the Minute Ring: Watch the Hour Hand Follow Automatically

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-LEFT)

1 EXPLORING CONTINUOUS EPICYCLIC MOTION

1. **ROTATE THE MINUTE RING BY HAND:**
Grip the outer knurled ring and rotate it clockwise.
Watch the hour hand move automatically through internal gears!

2. **TEST 1 FULL REVOLUTION (60 MINUTES):**
- Start both hands at 12:00.
- Turn minute ring 360° until minute hand lands back on 12.
- **Observe hour hand: It has advanced exactly to 1:00 (30°)!**

3. **DIGITAL LID PRACTICE:**
Drop hour block [1] and interval block [:00] into the 3x2 lid to bridge analog clock hands with digital displays (1:00 PM).

WHY PLANETARY GEARING MATTERS FOR LEARNING

Toy teaching clocks often have loose hands that students move independently, teaching them to place the hour hand incorrectly (e.g. putting the hour hand on 2 at 2:50). The planetary gear train enforces realistic continuous movement!

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR — [🕒] / [g][▸HMS] /

STEP 1 (SPIN TO 3:30):
Align both physical hands at 12:00, then spin the Green minute hand 3.5 turns to 3:30.

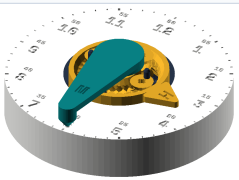
STEP 2 (CONVERT WITH [f][▸HR]):
Press 3.30 [f] [▸HR] → converts 3 hours 30 minutes into 3.5 decimal hours in Register X.

STEP 3 (MULTIPLY BY 30°/HR):
Press 30 [×] → X = 105 (proving the hour hand sits at 105°, halfway between 3 and 4)!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 105 → DISPLAY (X) = 105

LESSON 1: 12:1 PLANETARY CLOCK AT 3:30 ↔ [f][▸HR] KEY

STEP A: HANDS-ON WITH GEARED CLOCK HANDS — 1:1 CAD SCENE



Minute Hand: :30	Hour Hand: 3.5h	Gear Ratio
180° (Half Turn)	105° (Halfway 3→4)	12 : 1 Exact

▲ **When the minute hand spins 180° down to :30, the internal** planetary gears automatically advance the hour hand halfway to 4!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

12:1 GEAR RATIO TREE AT 3:30

3:30 → 3.5 HOURS

3.30 [f][▸HR] = 3.5 | × 30° = 105° Hr

3 Full Turns | + 180° (:30) | = 105° Hr

3.30 [f][▸HR] 30 [×] = 105°

PRESS BOX ICON & RPN KEYS (SCG):

1. TO DEC HR: 3.30 f ▸HR = 3.5 hr
2. HOUR ANGLE: 30 × = 105°

✓ DISPLAY (X) = 105

KEYSTROKES: 3.30 [f] [▸HR] 30 [×]

Spin the hands to 3:30, then convert 3.30 to 3.5 decimal hours with [f][▸HR] and multiply by 30°/hr!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 105 → X = 105

LINKED 12:1 ROTATION • 360° MINUTE TURN = 30° HOUR ADVANCE

CONTINUE RIGHT → PAGE 04

04 EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

PAGE / 08

Elapsed Time as Pie Wedges: Three 15-Minute Slices = 45 Minutes

01 02 03 04

SKU: TIME-01 • KEEP IN LID

SIDE A • 2ND UNFOLD (BOT-RIGHT)

2 ELAPSED TIME AS ANGULAR ARC ADDITION

THE PROBLEM: Find elapsed time from 2:15 PM to 3:45 PM.
Subtracting time with 60-minute bases confuses many students.
Instead, use physical angular arc rotation:

1. **SET CLOCK TO 2:15 PM:**
Hour hand is past 2; minute hand points to 3 (15 min).

2. **ADVANCE MINUTE RING TO 3:45 PM:**
- Rotate 1 full circle (360°) to reach 3:15 PM → 60 Minutes.
- Rotate another half circle (180°) from 3:15 to 3:45 → 30 Minutes.

Total Angular Turn = 360° + 180° = 540°

3. **DROP IN QUADRANT ARCS TO CONFIRM:**
Drop in 15-minute quadrant wedges: 4 wedges (60m) + 2 wedges (30m)
→ **Total Elapsed Time = 6 wedges × 15 min = 90 Minutes (1 hr 30 min)!**

✓ NO BASE-60 ARITHMETIC CONFUSION

Elapsed Time = Arc Swept by Minute Ring

540° / 360° = 1.5 Hours = 90 Minutes

Students literally see and feel the elapsed duration as an accumulation of physical quadrant wedges without having to carry or borrow minutes.

★ 2-PLAYER GAME: 'TIME TRAVELER RELAY' (2 MINS)

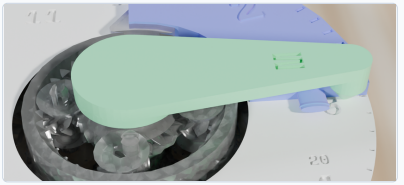
Player A sets a starting time on the geared clock (e.g., 3:30) and places 1 to 3 pie wedges beside the box. Player B predicts the ending time, lays the wedges onto the dial to verify, **and checks the answer on the calculator with [f][▸HR] and [g][▸HMS]!**

⚙ WHERE THIS SHOWS UP IN AUTOMOTIVE TRANSMISSIONS & ROBOTICS

Look inside the clear center window of your clock: that sun-planet-ring mechanism is an epicyclic planetary gearbox! The exact same gear geometry powers electric car drivetrains, **aircraft turbine reductions, and robotic arm joints.**

LESSON 2: 15-MIN PIE WEDGES ↔ [g][▸HMS] TIME ADDITION

STEP A: HANDS-ON WITH PIE WEDGES ON DIAL — 1:1 CAD SCENE



1 Wedge = 15m	2 Wedges = 30m	3 Wedges = 45m
1/4 Hr = 90°	1/2 Hr = 180°	3/4 Hr = 270°

▲ **Drop 15-minute pie wedges onto the clock face to see 3:30 + 0:45** = 4:15, then verify with [g][▸HMS]!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [🕒] / [g][▸HMS] /

ELAPSED TIME TREE: 3:30 + 0:45 = 4:15

3:30 + 3×[15m] WEDGES

3.5h + 0.75h = 4.25 | [g][▸HMS] = 4.15

+15m (3:45) | +15m (4:00) | +15m (4:15)

Bridges Across 12:00 Datum → 4:15!

PRESS BOX ICON & RPN KEYS (SCG):

1. ADD HOURS: 3.30 f ▸HR 0.45 f ▸HR +
2. TO H.MM: g ▸HMS = 4.15 (4:15)

✓ DISPLAY (X) = 4.15

KEYSTROKES: 3.30 [f] [▸HR] 0.45 [f] [▸HR] [+] [g] [▸HMS]

Lay three 15-minute wedges past 3:30 on the dial, then add in [f][▸HR] and convert back with [g][▸HMS]!

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 4.15 → X = 4.15

TIME AS FRACTIONS OF A CIRCLE • 15 MIN = 1/4 HOUR = 90°

STEP 3: FLIP SHEET OVER ↻ PAGE 05

05 — EXAMPLE 3 & DESK CHALLENGES

PAGE / 08

Quarter-Hour Wedges & 60-Minute Factoring + 3 Desk Challenges

05 06 07 08 — SKU: TIME-01 • KEEP IN LID — SIDE B • FLIP SIDE (TOP-LEFT)

3 QUARTER-HOUR WEDGES & 60-MINUTE FACTORING

THE GOAL: Show how 60 minutes factors into 4 clean 15-minute quadrant wedges!

1. INSERT 15-MIN WEDGE (:15):
Drop the 90° colored quadrant wedge into the 12:00 to 3:00 dial sector.

2. ADVANCE THE MINUTE HAND:
Rotate the outer ring 90°—the minute hand traces the wedge edge cleanly to :15.

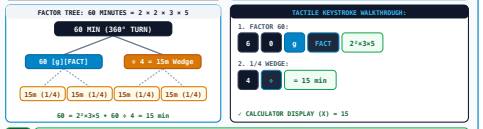
3. SNAP DIGITAL BLOCKS IN LID:
Seat [03], [:15], and [PM] into the lid sockets to link analog and digital notation.

✓ **60 IS A HIGHLY COMPOSITE NUMBER: $2^2 \times 3 \times 5$**

An hour has 60 minutes because 60 has 12 factors: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.
Four 15-min wedges fill 360°: 60 / 4 = 15 min (90° each) with zero fractions!
Synchronized 12:1 planetary gears advance the hour hand 1/4 of an hour (7.5°).
Verify on calculator: 60 [g][FACT] 4 [÷] confirms 15 min per quarter-hour wedge!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER
Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE



STEP B: PRESS BOX ICON & RPN KEYS (SC6)

FULL KEYSTROKES: 60 [g] [FACT] 4 [÷]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 15 → X = 15

3 DESK CHALLENGES — TRY THESE NOW!

WARM-UP #1 — **Set the clock to 4:30. What angle is between 12 and the minute hand?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #1 below!

LEVEL-UP #2 — **How many 15-minute quadrant wedges fit between 1:00 and 2:30?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #2 below!

BRAIN-MELTER #3 — **Spin the minute hand 2 full turns (720°). How far did the hour hand move?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#3: 2 hours (60°)
#2: 6 wedges (90 minutes)
#1: 180° (1/2 circle) | #2: 6 wedges (90 minutes)

FOR PARENTS & TEACHERS (Grades K–4 (Time, Elapsed Intervals, Fractions & Angles))

- CCSS.MATH.CONTENT.1.MD.B.3: Tell and write time in hours and half-hours.
- CCSS.MATH.CONTENT.3.MD.A.1: Tell and write time to the nearest minute & measure intervals.
- CCSS.MATH.CONTENT.4.MD.C.5: Recognize angles as geometric shapes formed by two rays.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 — **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([⬚] / [g][•HMS] / [f][•HR]) and keystrokes on your calculator:
• **60 [g] [FACT] 4 [÷] → X = 15**

CHECK OFF YOUR MASTERY MILESTONES:

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

06 — HOW TO STACK IT ALL BACK IN

PAGE / 08

The 10-Second Self-Checking Pack-Up Puzzle

05 06 07 08 — SKU: TIME-01 • KEEP IN LID — SIDE B • PACK-UP GUIDE (TOP-RIGHT)

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:



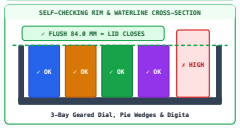
✓ **CUSTOM CIRCULAR CRADLE & WEDGE BAYS**

Drop the geared clock dial into the large circular cradle on the left.
Stack the 4 quarter-hour pie wedges into the middle bay, and snap the 6 digital minute blocks (:00 to :45) and AM/PM tags into the right-hand organizer slots.
One quick glance confirms all 19 pieces are present before you close the lid!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE



▲ **SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:**
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

DIGITAL DISPLAY LID & GEAR CARE

- The top of the lid has recessed sockets for building digital times (HH : MM AM/PM).
- Dry-running gears: never add oil or grease (it attracts dust). Just rinse in cool water if needed.
- Turn the minute ring to move the hands—never force the slow hour hand backward!

SHORT A PIECE AT CLEAN-UP TIME?
First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ TIME-01 4-POINT CLEAN-UP CHECKLIST

1. CRADLE CLOCK CORE:	Seat the geared clock dial flat into the left circular bay.
2. STACK 4 WEDGES:	Place all 4 quarter-hour pie wedges (Q1–Q4) into the center bay.
3. SEAT DIGITAL TILES:	Snap the 6 minute tiles (:00..:45) & AM/PM blocks into the right slots.
4. CLEAR DIGITAL LID:	Remove any tiles from the top lid sockets & snap the 3×2 lid shut!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

MOLDED CLOCK CRADLE •DEDICATED BAYS FOR ALL 19 PIECES

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

07 — THE STACKCALC CONNECTED LAB

PAGE / 08

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08 — SKU: TIME-01 • KEEP IN LID — SIDE B • ECOSYSTEM (BOT-LEFT)

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS
Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 Ten-Frames Lab — Early Counting & Regrouping	**DOMI-01** Math Domino Studio — Doubles & 12-Frame Part-Whole	**FRAC-01** Fraction Studio — Proportional Towers & 4-Way Faces
PRIME-01 Prime Factor Studio — Multiply by Stacking Prime Towers	**ALGB-01** Algebra Studio — Quadratic Area Arrays & Factoring	**TIME-01** ◂ YOUR BOX — Planetary Clock — 12:1 Geared Hands & Time Wedges
PROB-01 Probability Dice — Fair Polyhedrals & 10-Sector Spinner	**VOLV-01** Unit Circle Volvelle — Spin-to-Read Exact Trig Radicals	**SC-32-HW** SC-32 RPN Calculator — 4-Register Stack Scientific Engine

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH
In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. TIME-01 BOX LID ICON



2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN



▲ **BOX ICON ↔ CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- Time interval blocks (:00..:45) snap into the same 19.6mm sockets used across all StackCalc kits.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Every hour mark is 30° (π/6 rad)—connecting early clock reading directly to the Unit Circle Volvelle!
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS
Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.
Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK



HOW BLOCKS MAP TO STACK KEYS

1. **PLACE 1ST PIECE ON STAGE:** Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:** Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:** Press [⬚] / [g][•HMS] / [f][
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [⬚] / [g][•HMS] / [f][•HR] • VERIFIED LESSON SEQUENCES:
• Lesson 1: 3.30 [f] [•HR] 30 [×] → X = 105
• Lesson 2: 3.30 [f] [•HR] 0.45 [f] [•HR] [+] [g] [•HM → X = 4.15

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

08 — KEEP THIS CARD! PART IDs & VIDEO

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Complete 19-Piece Reference & 60-Second 3D Video Guide

05 06 07 08 — SKU: TIME-01 • KEEP IN LID — SIDE B • PART REFERENCE (BOT-RIGHT)

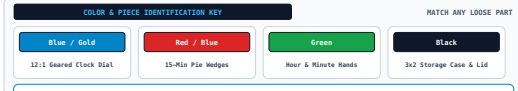
PART ID REFERENCE TABLE (19 TOTAL PIECES)

✓ **KEEP THIS CARD INSIDE YOUR BOX LID!**
Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

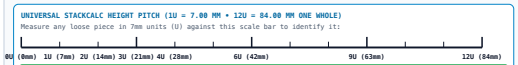
PART NAME	QTY	COLOR / FINISH	PART ID
12:1 Planetary Clock Dial	1x	Blue / Gold	TIME-CLOC-db6
15-Min Pie Wedges (Q1–Q4)	4x	Red / Blue	TIME-CLOC-db6
Minute Blocks (:00 to :45)	6x	Blue	TIME-CLOC-1f0
AM / PM & Hour Blocks	6x	Gold / Red	TIME-CLOC-aa0
3x2 Storage Case & Lid	2x	Black	TIME-CLOC-db6

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at stackcalc.io/parts to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY — MATCH ANY LOOSE PART



UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU TIME-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/classroom-clock/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 5 (Geared Time & Angles) • SC-32 Manual Ch. 4
STANDARDS: CCSS 1.MD.B.3, 2.MD.C.7, 3.MD.A.1, 4.MD.C.5 • TEKS 2.6G, 3.7C (Time & Angles)
WATERLINE SPEC: 3-Bay Geared Dial, Pie Wedges & Digital Tile Deck

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):
Ex 1: 3.30 [f] [•HR] 30 [×] → X=105
Ex 2: 3.30 [f] [•HR] 0.45 [f] [•HR] [+] [g] [•HMS] → X=4.15
Ex 3: 60 [g] [FACT] 4 [÷] → X=15

BOX ICON KEY: [⬚] / [g][•HMS] / [f][•HR]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!